Hi,

p $\wedge$ q

$\wedge$pq

$p \wedge q$

$\wedge pq$

$\sim$

$\sim$p

The set $\mathbb{N}$ contains all natural numbers. For any $x \in \mathbb{R}$, we have $x^2 \geq 0$. The complex numbers $\mathbb{C}$ extend the reals.

1. Prove the following directly from the two definitions and three axioms given in ICA2 on 8/20. You may not use any other results without proving them first. Every statement in your proof must be fully justified. See **Elements of (Proof) Style** guidelines in Canvas for information.

   (a) Direct proof: For all integers n, if n is odd, then $9n + 5$ is even.

   (b) Contrapositive proof: For all integers n, if $7n + 5$ is odd, then n is even.

   (c) Proof by contradiction: For all integers n, if $3n + 2$ is even, then n is even.